

AUTO TICKET CLASSIFICATION USING FLOW DESIGNER

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1.OBJECTIVE

Automatically classify incidents at the time of creation

- Reduce manual effort for IT agents
- Improve ticket routing efficiency
- Implement a no-code, easily maintainable solution

2. ARCHITECTURE DIAGRAM



3. SYSTEM REQUIREMENTS

3.1. SOFTWARE REQUIREMENTS

Software requirements specify the ServiceNow platform, Flow Designer, Incident Management, web browser, and internet connection required to automatically classify and assign support tickets.

1. Software requirements specify the software tools and platforms needed to develop and run the system.
2. For Auto Ticket Classification, ServiceNow, Flow Designer, Incident Management, and a web browser are required.

3.2. HARDWARE REQUIREMENTS

Hardware requirements are the physical devices and system resources needed to develop, access, and run the Auto Ticket Classification using Flow Designer system.

1. Computer/Laptop – Used to access ServiceNow and configure the Flow Designer.
2. Internet connection & basic system resources – Required for smooth access to the ServiceNow cloud platform.

4. IMPLEMENTATION

PHASE 1: REQUIREMENT ANALYSIS & PLANNING

The school IT helpdesk receives multiple support request daily from students and teachers related to common issues such as network connectivity, hardware failures, account access problems, and system performance issues.

Currently, IT staff manually review each ticket to:

- Identify the issue type

- Select the correct category and
- Inform the caller about ticket creation.

This manual process is time-consuming, error-prone, and not scalable as the number of requests increases.

Activity : 1 Business Use Case.

Business Requirements

The system must:

Automatically classify IT tickets based on the issue description. Assign both Category and Subcategory without manual intervention. Support dependent choice logic between Category and Subcategory. Send an automated email notification to the caller upon ticket creation Store ticket information in a structured and standardized format. Ensure easy maintenance and future scalability.

Activity : 2 Creation of New Update Set.

Navigation Steps :

- Go to All menus → Search for Update set.
- Under System Update Sets → Select Local Update Sets.
- Click the New button on the top right corner of the page.

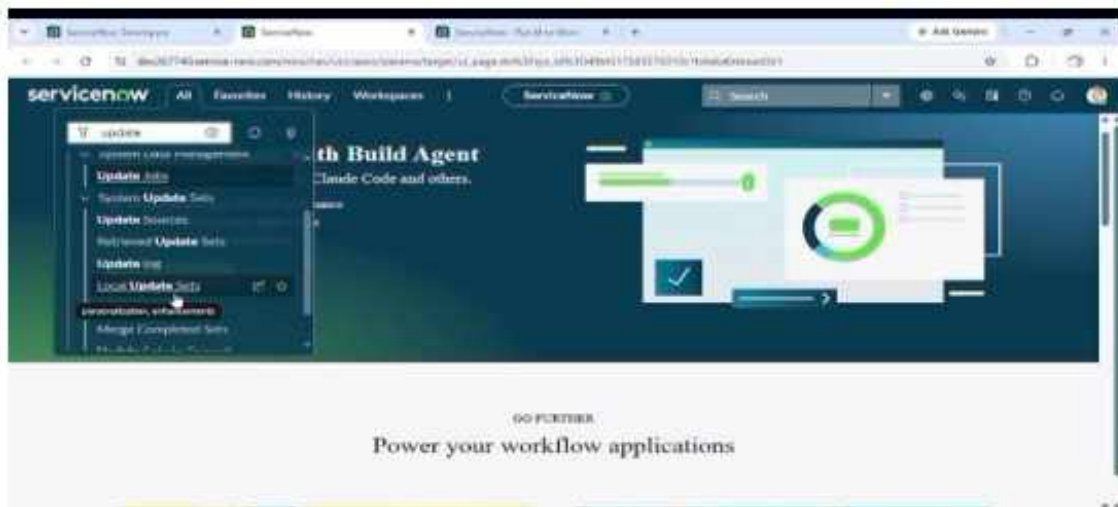
Name : Project Update Set.

State : In progress.

Application : Global.

- Then Save the Form.
- Under Related links Click on the Make This My Current Set.
- Then the Created Update Set will appear as Current.

Activity: 3 Navigation steps in Picture View



The screenshot displays the 'Update Sets' page in the ServiceNow interface. The top navigation bar includes the 'servicenow' logo and various utility buttons. The main content area features a search bar and a table of update sets. The table has the following columns: Name, Application, State, Installed from, Created, Created by, Parent, and Batch Base. Two update sets are listed: 'Default' and 'Project Update Set'. The 'Project Update Set' is currently selected, highlighted in blue. Below the table, a 'Related Links' section contains a link to 'View Update Sets'.

Name	Application	State	Installed from	Created	Created by	Parent	Batch Base
Default	Global	In progress		2026-06-12 14:20:10	system	(empty)	(empty)
Project Update Set	Global	In progress		2026-09-22 00:27:07	admin	(empty)	(empty)

Related Links

[View Update Sets](#)

ServiceNow

Update Set - Create Project Update ...

Update Set
New record

Name: Project Update Set 2

State: In progress

Application: Global

Parent:

Release date:

Description:

Submit Submit and Make Current

Phase :- 2 Backend Development & Configuration

Activity 1 : Creation of Custom Table to Store Ticket Records.

Steps to Create Custom Table

- Open All menus → Search for Tables.
- Under System Definition → Click Tables.
- On the top right corner Click New button.

Label : Incident WorkFlow.

- Uncheck the Create module.
- Now **Save** the Form.

Then Click the Controls Suction and Enable the Check box true Auto Number.

Now again save the Form.

servicenow

Table

System Classic Mobile UI

Table Titles

System Definition

Table

Create and manage tables

Table Instances

Custom Tables

Remote Tables

Table

State	Installed from	Created	Created by	Parent	Batch Base
In progress		2026-06-12 14:20:10	system	empty	empty
In progress		2026-07-22 00:27:07	admin	empty	empty
In progress		2026-07-22 00:40:40	admin	empty	empty

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Tables

Update Name is not empty

Label	Name	Extends table	Extensible	Updated
Account Subscription Entitlement	account_subscription_entitlement	empty	false	2026-06-12 14:52:23
Adaptive Authentication Event	adaptive_auth_event	empty	false	2026-06-12 14:41:06
Agent Asset Recommendation	agent_asset_recommendation	Application File	false	2026-06-12 14:55:01
MDI Server File	agent_file	empty	false	2026-06-12 14:22:34
AI/ Access Policy	alg_access_policy	Application File	false	2026-06-12 14:01:27
AI/ Access Policy Scope Mapping	alg_access_policy_scope_mapping	Application File	false	2026-06-12 14:01:28
AI/ Scope Tool Mapping	alg_scope_tool_mapping	Application File	false	2026-06-12 14:01:26
Record Producer Configuration	als_rp_config	Application File	false	2026-06-12 14:39:11
Search Actions	als_ui_action	Application File	false	2026-06-12 14:39:11
AI/ Search ACL Overrides	als_acl_overrides	Application File	false	2026-06-12 14:03:37
AI/ Search Active Table Ingestion Tracker	als_active_table_ingestion_tracker	empty	false	2026-06-12 14:03:35
AI/ API Transaction Log	als_api_transaction_log	empty	false	2026-06-12 14:03:39

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Table - New Record

ServiceNow recommends creating custom tables in scoped applications. To learn more about creating scoped applications, click [here](#).

A table is a collection of records in the database. Each record corresponds to a row in a table, and each field on a record corresponds to a column in that table. Applications use tables and records to manage data and processes. [More info](#)

* Label: Incident Workflow

* Name: ai_incident_workflow

Extends table:

Application: Global

Create module

Remote Table

Columns Controls Application Access

Then **Save** the Form and Under Related Links Click **Show Form**.

Now Open the Configure On clicking the Context Menu and then **Open** the Form Designer.

Then Create Fields as Required.

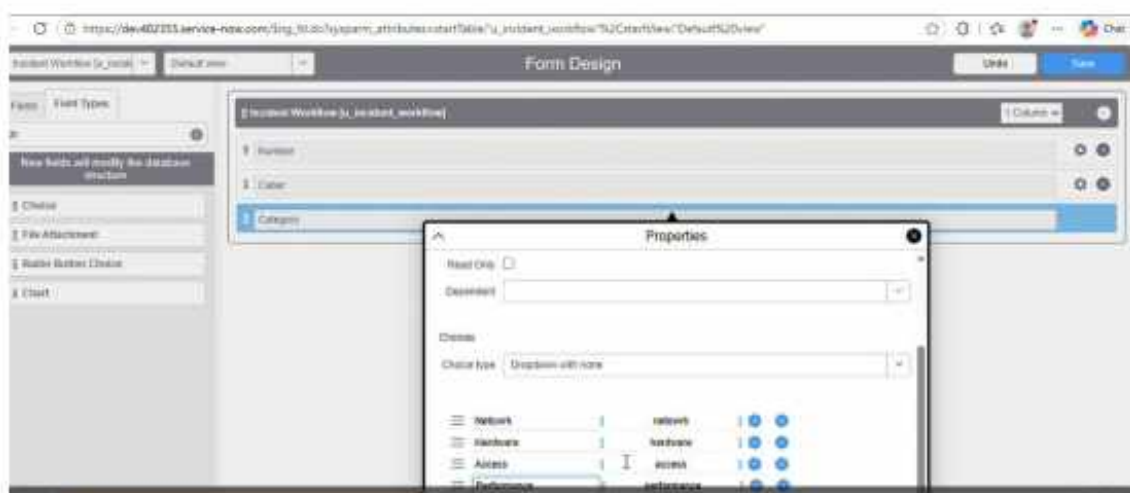
Activity 2: Field Creation and Data Type Configuration

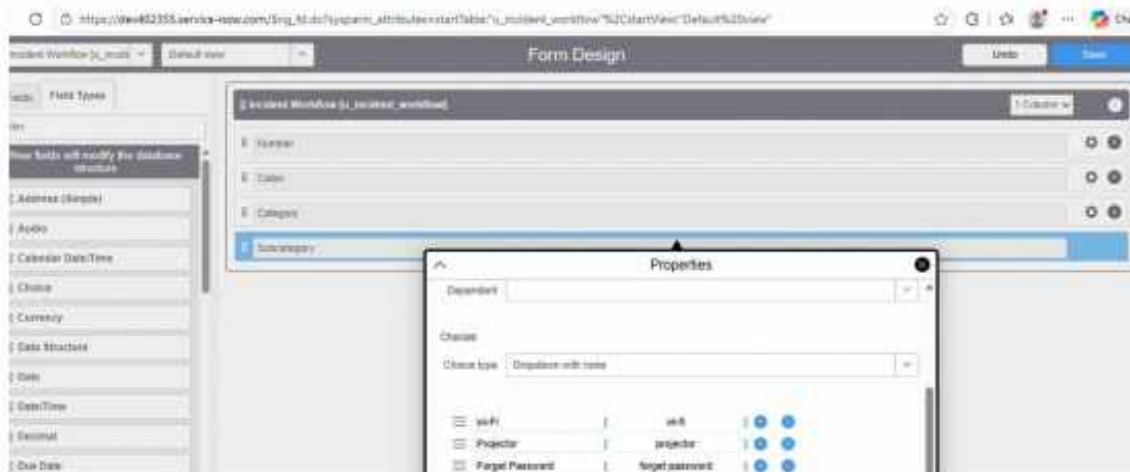
The objective of this milestone is to create all required fields in the School IT Ticket table with correct data types, choices, and dependencies, ensuring structured and consistent data capture.

Field Label	Type	Reference/Choice
Number	Auto Number	-
Caller	Reference	sys_user
Category	Choice	[Network,Hardware,Access,Performance]
Subcategory	Choice	- [New ,In progress ,On hold, Resolved, Closed]
Short Description	String	--
Description	String	--
State	Choice	- [New ,In progress ,On hold, Resolved, Closed]
Assigned group	Reference	Sys_user_group
Assigned to	Reference	Sys_user

After Creating Fields with their appropriate Type, then add the Choice from the Form design.

- To all the Choice list Fields.
- They are Category, Subcategory and State.
- Same like that, add Choice to the State field.
- Then Click the save button on the top right corner.
- Open the Form and reload it.





- Same like that, add Choice to the State field.
- Then Click the save button on the top right corner.
- Open the Form and reload it.

Activity 3 : Implementing Dependency Between Category and Subcategory Choice Fields

The objective of this milestone is to implement a dependency relationship between the Category and Subcategory fields so that the available Subcategory options dynamically changes based on the selected Category.

Navigation Steps :

- ❖ Open the Created Table Form and Right click on the **Subcategory** Field.
- ❖ Click on the Configure Dictionary.
- ❖ Under **Related links** Click on the Advanced view,
- ❖ Then select the Dependent Field Section.
- ❖ Enable the checkbox Use dependent field true.
- ❖ Then select the Field Category on the Dependent on field.

Then **save** the Form.

Dictionary Entry - Subcategory

Allows the behavior of a field or functionality that depends on the field. [More info](#)

Attributes:

Choice List Specification | Dependent Field | Calculated Value | Default Value

Specify a field whose value is used to determine choice list options for this field. [More info](#)

Use dependent field ☒

Dependent on field:

Create Choice List | Delete Column | Update

Related Links

[View Table](#)

[Run Report](#)

Access Controls | Choices (0) | Attributes (1) | Labels (1)

Now scroll down to the choice section.

Then add the dependent fields for which option dependent on the category gives the category choice backend name to the subcategory choice values.

Wi-Fi → **Network.**

Projector → **Hardware.**

Forgot password → **Access.**

Slow computer → **Performance.**

Dictionary Entry - Subcategory

Dependent on field: Category

Related Links: Show Table, Run Report Scan

Access Controls: Choices (4), Attributes (2), Labels (2)

Label	Value	Language	Sequence	Inactive	Updated	Dependent value
WiFi	wi-fi	en	0	False	2026-09-23 00:44:14	
Slow computer	slow-computer	en	3	False	2026-09-23 00:44:14	
Projector	projector	en	1	False	2026-09-23 00:44:14	
Forgot Password	forgot password	en	2	False	2026-09-23 00:44:14	

1 to 4 of 4

Then **save** the Form.

Then it will display each choice to each one separately based on the category subcategory displays.

Milestone 3 : Automation using Flow Designer & Email Notification.

When a student creates a new IT support ticket, they only need to enter the Caller Name and Short Description of the issue

- ❖ If the short description contains the word WiFi, the flow assigns Category = Network and Subcategory = WiFi.
- ❖ If the description contains Projector, the flow assigns Category = Hardware and Subcategory = Projector.
- ❖ If the description includes Password or Login, the flow assigns Category = Access and Subcategory = Forgot Password.

If the description mentions Slow or Hanging, the flow assigns **Category = Performance** and **Subcategory = Slow Computer**.

Activity : 1 Auto Classification using Flow design.

- Go to All menus → **Search Flow designer.**
- Under Process Automation **Open the Flow designer.**
- On the top right corner **Click New** and Select Flow Option.

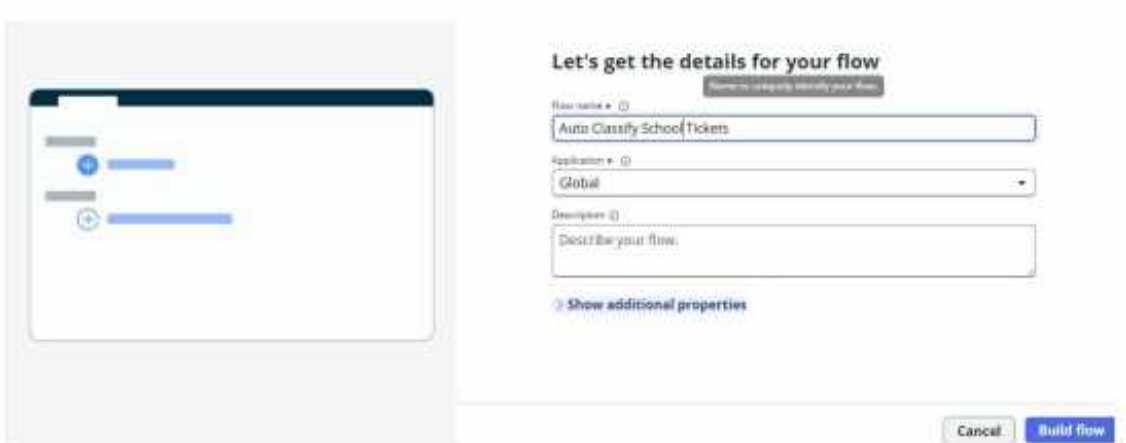
Name : Auto Classify School IT Tickets.

Application : Global.

- Now Click the Build Flow button.



State	Installed from	Created	Created by	Parent	Search Name
In progress		2020-08-14 10:30	system	empty	empty
Complete		2020-08-14 10:30	admin	empty	empty



Let's get the details for your flow

Flow name:

Application:

Description:

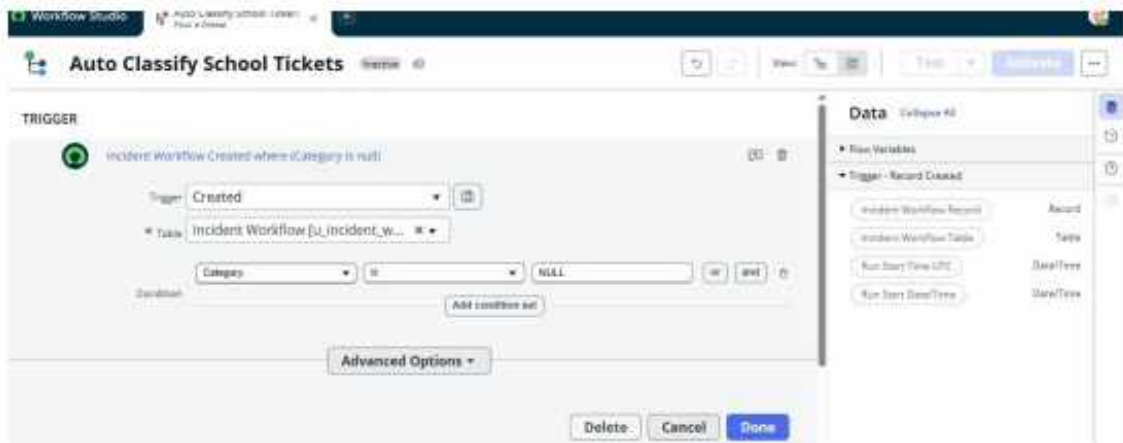
[Show additional properties](#)

Activity : 2 Trigger Configuration.

Trigger : Record Created

Table : Incident Workflow.

Condition : Category is Empty. Then Click Done.



Activity : 3 Wi-Fi Issue Classification.

Select **If** condition

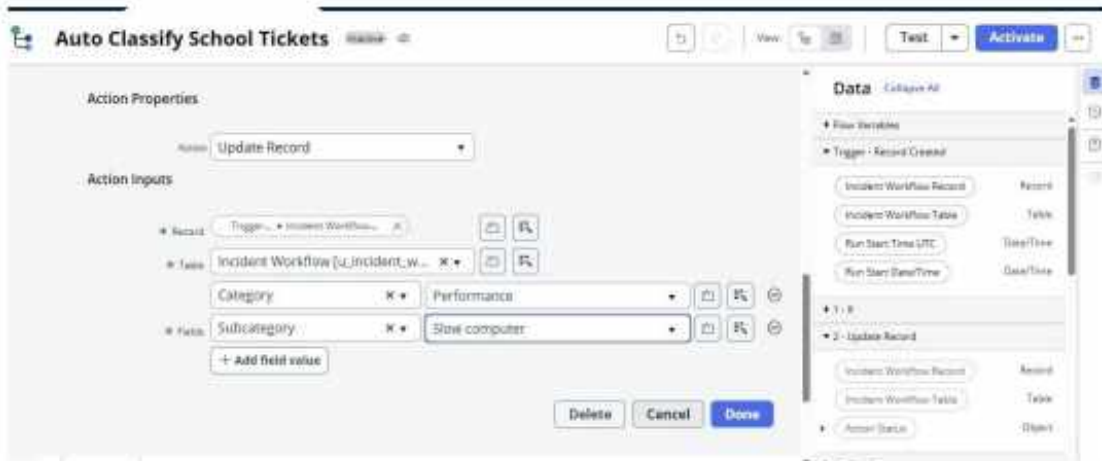
Condition label : SD is Wifi or Network (Give as ur wish).

Condition 1 : Using data pill picker Select Short description as Field , and operator Contains , Value is Wi-Fi.

Condition 2 : Using data pill picker Select Short description as Field, and Operator



Fields Set **Category** = Network.Set **Subcategory** = Wi-



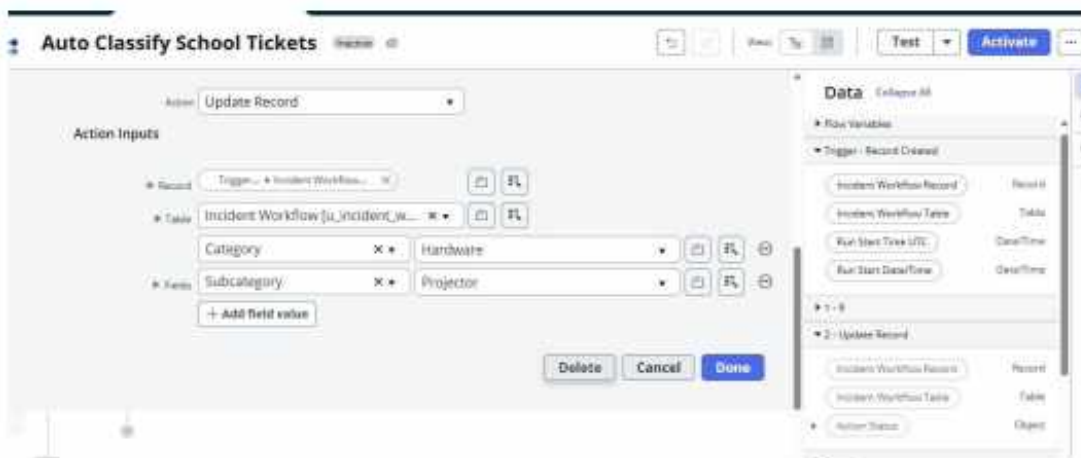
Activity : 4 Projector Issue Classification.

- Now Select the **Else If Condition**.

Condition label : SD is Projector or Hardware.

Condition : Short description Contains Projector.

- Then Click the Done button. Now Action is an Update record.



Activity : 6 Slow Computer Classification.

- Now Again Click the **Else If Condition**.

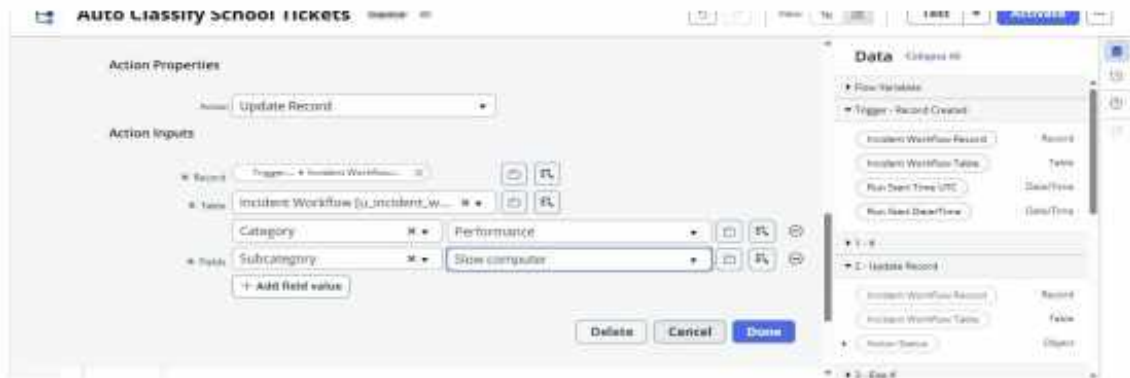
Condition label : SD is Performance.

Condition : Short description contains Slow Computer.

- Then Click the Done button. Now Action is an Update Record.

Set **Category = Performance**.

Set **Subcategory = Slow computer**. Then Click the Done button.



Activity : 7 Email Action Setup.

After classification, an email notification is sent using Flow Designer.

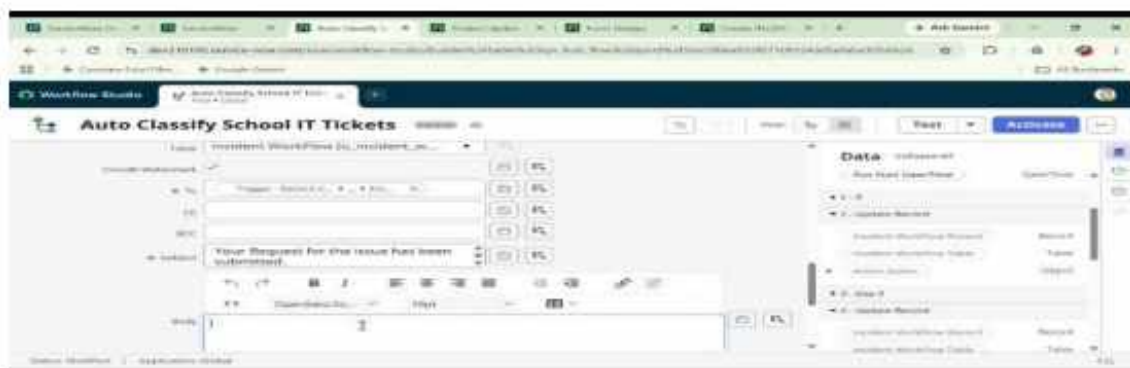
Set Action Send Email.

To : Caller → Email.

Subject : Your Request for the issue has been submitted.

Body : Ticket confirmation message.

Then Click the Done button.



PHASE 6 : Testing, Validation & Security.

The objective of **Phase 4** is to ensure that the developed solution works as expected, meets all functional requirements, and follows basic security and access control best practices.

This phase validates the end-to-end automation, confirms data accuracy, and ensures that only authorized users can access or modify ticket data

Data Validation Checks

1. Mandatory fields captured correctly.
2. Auto-number generated without duplication.
3. Category and Subcategory values stored accurately.
4. Reference fields resolve correctly to user and group tables.

Activity : 1 Testing & Validation.

- Test Scenario 1
- Enter one Caller Name.
- Short Description: WiFi not working in library
- Then Save the Form,
- And Reload the Form.
- Then it fills auto as :
Category → Network.
Subcategory → WiFi.
- Email sent to caller Email Id

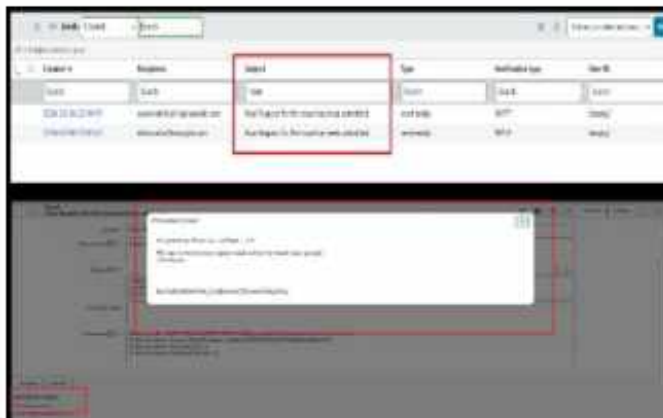


The screenshot shows a web-based form for creating a ticket. The form is titled 'New Ticket - Description' and 'New Incident'. It contains several dropdown menus and text fields. The 'Module' dropdown is set to 'Email'. The 'Category' dropdown is set to 'Network'. The 'Subcategory' dropdown is set to 'WiFi'. The 'Description' text field contains the text 'WiFi not working in library. Then save the form'. The 'Priority' dropdown is set to 'Medium'. The 'Date' field is set to '09/14/2025'. The 'Reopened' checkbox is checked. At the bottom, there are links for 'New Ticket' and 'My Profile'.

Activity : 2 Check Email Notification.

- Open All menus → Search Emails.

- Under System logs → Select Emails.
- Search with the Given Subject Name.
- Open that Record and Under Related Links Click Preview mail Option.



Activity : 3 Test Scenario 2

- Enter one Caller Name.
- Short Description: Projector not turning on.
- Then Save the Form.
- And Reload the Form.
- Then it fills auto as :
Category → Hardware.
Subcategory → Projector.
- Then Email sent to caller Mail id.

PHASE 5: Deployment & Final Documentation.

The objective of Phase 4 is to deploy the developed solution to the target environment and finalize all project documentation.

This phase ensures that the solution is stable, reusable, and ready for demonstration or production use, along with complete and clear documentation for future reference.

Activity : 1 Make Update set to Complete State.

- Go to All menus → Search For Local Update Sets.
- Under System Update sets Select → Local Update sets.
- Open the Project Update set.
- Make the State From In progress to Complete.
- Then Save the Form.
- Under Related links Click Export to XML Option.
- Then the Xml Format the Update set data will be downloaded to your system.
- We can share through WhatsApp or Mail that downloaded file.
- Then the Created Update set will transform to others in this way.



Conclusion

- The Auto Ticket Classification using Flow Designer project successfully delivers a complete, end-to-end automation solution for a school IT helpdesk. The primary objective of reducing manual effort in ticket categorization and improving response efficiency has been fully achieved.
- Through structured requirement analysis, careful backend configuration, and no-code automation using Flow Designer, the system now automatically classifies IT tickets based on the short description provided by students. The implementation of dependent choice fields ensures data accuracy, while automated email notifications improve communication by instantly acknowledging ticket creation to the caller.
- The solution was developed using a phased approach, ensuring clarity, stability, and ease of maintenance. Comprehensive testing and security validation confirmed that the system functions reliably, maintains data integrity, and enforces appropriate access controls.
- Overall, this project demonstrates how Service Now can be effectively used to build scalable, maintainable, and user-friendly automation solutions without relying on complex scripting or machine learning. The design is well-suited for

small to medium environments such as educational institutions and can be easily extended in the future to include advanced features like assignment automation, SLA tracking, or predictive intelligence.